

Explainability-Guided SegFormer for Forest Cover Segmentation Using Attention Consistency Supervision

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Abstract

Vision Transformer (ViT)-based segmentation models achieve strong accuracy, but their internal attention is typically used only for post-hoc visualization, not as a supervised training signal. Prior work applying attention supervision for interpretability has focused on medical imaging and autonomous-driving domains, while forest/non-forest segmentation from aerial imagery still treats attention purely as an architectural feature-refinement mechanism, with interpretability evaluated qualitatively if at all. We present an explainability-guided SegFormer-B0 architecture that treats attention as a first-class training target: an *Attention Consistency Loss* aligns the model’s internal attention with the ground-truth forest mask during training, encouraging the network to focus on canopy regions rather than roads, shadows, or built structures. Because SegFormer’s encoder uses spatial-reduction attention rather than standard multi-head self-attention, we adapt Gradient-weighted Attention Rollout to SegFormer, restricting it to the final encoder stage where attention is square and token-consistent. We introduce a quantitative metric, Average Attention–Mask Overlap (AAMO), to evaluate interpretability numerically rather than only visually. In preliminary small-scale experiments, the proposed method raises AAMO from 0.0144 to 0.432 relative to a vanilla SegFormer-B0 baseline while also slightly improving Dice (0.8017→0.8191) and IoU (0.6690→0.6936), alongside a U-Net baseline (Dice 0.8492). Full-scale validation and a complete ablation study are ongoing.

CCS Concepts

• **Computing methodologies** → **Neural networks; Image segmentation; Interpretability and explainability.**

Keywords

semantic segmentation, vision transformers, SegFormer, attention supervision, explainable AI, forest cover mapping, remote sensing

1 Introduction

Semantic segmentation of forest cover from aerial imagery underpins ecological monitoring, deforestation tracking, and land-use management. Transformer-based models such as SegFormer [9] have been applied to this task for their ability to capture long-range spatial dependencies, but the great majority of transformer segmentation work, including recent forest-specific studies [1, 5, 7], uses attention purely as an architectural mechanism to improve accuracy, evaluating explainability only through post-hoc qualitative heatmap inspection, if at all.

Transformer attention in segmentation models is not explicitly guided during training and frequently attends to task-irrelevant

regions — roads, shadows, buildings — rather than the class of interest, reducing both accuracy at ambiguous boundaries and the trustworthiness of the model’s explanations. No existing work addresses this specifically for forest/non-forest segmentation, nor evaluates attention faithfulness quantitatively in this domain.

This paper makes four contributions: (1) an Attention Consistency Loss that supervises SegFormer-B0’s internal attention against the ground-truth forest mask during training; (2) an adaptation of Gradient-weighted Attention Rollout to SegFormer’s spatial-reduction attention; (3) AAMO, a quantitative interpretability metric evaluable alongside standard segmentation metrics; and (4) a preliminary empirical comparison against U-Net and vanilla SegFormer-B0 baselines showing that explainability-guided training does not cost — and may slightly improve — segmentation accuracy while substantially improving attention faithfulness.

2 Related Work

Forest-cover segmentation from aerial and satellite imagery has been approached almost exclusively as an accuracy problem. U-Net [8] remains the dominant convolutional baseline, and recent forest-specific datasets such as the WildUAV forest-inspection benchmark [2] have focused effort on acquiring larger and more varied labeled imagery (real and simulated, multiple altitudes and illumination conditions) to improve segmentation accuracy on models such as HRNet and PointFlow, without examining what the underlying network attends to or why. Existing work directly relevant to attention supervision falls into two families.

Attention-as-architecture. ForResANeXt [5] embeds lightweight attention in residual blocks for forest/non-forest satellite segmentation; Attention-Refined PP-LiteSeg [7] fuses multi-branch attention for cross-regional UAV forest segmentation; an attention-based U-Net [1] adds attention gates for deforestation mapping on Sentinel-2 imagery. All three use attention purely as a feature-refinement mechanism improving accuracy — attention is never supervised or evaluated as an explanation.

Attention-as-explainability. eX-ViT [10] regularizes ViT attention with an attribute-guided loss for weakly supervised natural-image classification; TransAttUnet [4] integrates guided attention into a U-Net/Transformer hybrid for medical segmentation; work on driving models [6] applies an explicit attention loss to align attention with task-relevant scene regions. All three supervise attention as a training signal, but outside the forest/remote-sensing domain and without a quantitative attention-mask overlap metric. The attention-extraction method underlying our approach is Chefer et al.’s Gradient-weighted Attention Rollout [3], originally defined for standard multi-head self-attention.

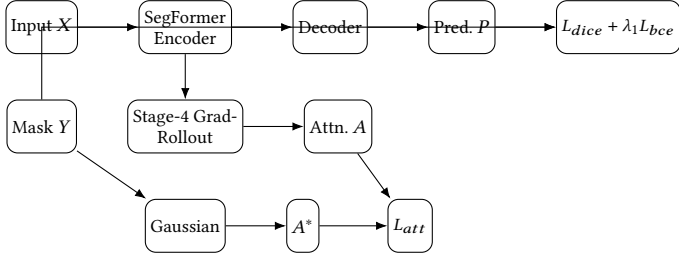


Figure 1: Training pipeline: predicted mask P and attention map A are each supervised, then combined into the total training objective L .

Gap. No prior work combines (a) attention supervision as a training signal, (b) the forest/non-forest remote-sensing domain, and (c) a quantitative interpretability metric evaluated alongside standard segmentation metrics. The attention-as-architecture literature [1, 5, 7] operates in the correct domain but never supervises or measures attention as an explanation. The attention-as-explainability literature [4, 6, 10] supervises attention as a training signal but never in the forest-segmentation domain, and none of the three reports a quantitative attention-fidelity metric comparable across model configurations. This paper addresses that combined gap: an Attention Consistency Loss that supervises SegFormer’s internal attention against the ground-truth forest mask, an adaptation of Gradient-weighted Attention Rollout to SegFormer’s spatial-reduction attention, and AAMO, a new quantitative metric that lets attention fidelity be reported and compared the same way Dice or IoU are. We also draw on the WildUAV forest inspection dataset work [2] as evidence that forest-aerial-segmentation research so far has focused effort on acquiring larger and more varied labeled imagery rather than on what the resulting models attend to.

3 Proposed Method

3.1 Architecture Overview

The pipeline (Figure 1) extends SegFormer-B0 with an explainability-supervision path applied during training only; at inference the model runs as standard SegFormer-B0 with negligible overhead. The encoder’s stage-4 attention (the only stage with spatial-reduction ratio = 1, giving square, token-consistent 64×64 self-attention) feeds an adapted Gradient-weighted Attention Rollout to produce attention map A . The ground-truth mask Y is Gaussian-smoothed into a soft target A^* . A and A^* are compared by the Attention Consistency Loss and combined with the standard segmentation loss.

3.2 Handling Spatial-Reduction Attention

Standard Attention Rollout assumes full multi-head self-attention across all tokens. SegFormer’s Efficient Self-Attention reduces keys/values by a spatial-reduction ratio at earlier stages ($sr = 8, 4, 2$ for stages 1–3), making those stages’ attention matrices rectangular rather than square. Only stage 4 ($sr = 1$) has true square, token-consistent 64×64 attention — the only stage where Rollout’s recursive matrix product is valid without further approximation, verified empirically on MiT-B0 at 256×256 input resolution.

Algorithm 1 One training step of the attention-consistency variant

Require: image X , ground-truth mask Y , model (E, D) , weights λ_1, λ_2

- 1: $F \leftarrow E(X)$ ▷ encoder features, all 4 stages
- 2: $P \leftarrow D(F)$ ▷ decoder segmentation output
- 3: $A \leftarrow \text{Rollout}(F_4)$ ▷ stage-4 attention only, autograd-attached
- 4: $A^* \leftarrow \text{Gaussian}(Y, \sigma=8)$ ▷ soft attention target
- 5: $L \leftarrow L_{\text{dice}}(P, Y) + \lambda_1 L_{\text{bce}}(P, Y) + \lambda_2 L_{\text{att}}(A, A^*)$
- 6: $L.\text{backward}(\text{create_graph}=\text{True})$ ▷ double backward: A depends on ∇F
- 7: $\text{optimizer.step}()$

3.3 Mathematical Formulation

Let $X \in \mathbb{R}^{256 \times 256 \times 3}$ and ground-truth mask $Y \in \{0, 1\}^{256 \times 256}$:

$$F = E(X), \quad P = D(F) \in [0, 1]^{256 \times 256} \quad (1)$$

$$A = \text{Rollout}(F) \in [0, 1]^{256 \times 256} \quad (2)$$

$$A^* = \text{Gaussian}(Y) \quad (3)$$

$$L_{\text{dice}} = 1 - \frac{2|P \cap Y|}{|P| + |Y|}, \quad L_{\text{bce}} = - \sum [Y \log P + (1-Y) \log(1-P)] \quad (4)$$

$$L_{\text{att}} = \text{MSE}(A, A^*) \text{ or } \text{KL}(A \parallel A^*) \quad (5)$$

$$L = L_{\text{dice}} + \lambda_1 L_{\text{bce}} + \lambda_2 L_{\text{att}} \quad (6)$$

with $\lambda_1=1, \lambda_2=0.3$ initially, tuned on the validation set. Because A is itself defined via a gradient (Grad-Rollout is Grad-CAM-style), back-propagating L_{att} requires double backpropagation; our implementation keeps A attached to the autograd graph via `create_graph=True` so one combined `backward()` reaches all parameters.

Algorithm 1 summarizes one training step. At inference, only lines 1–2 run — the Rollout/attention path is training-only supervision, so deployment cost equals vanilla SegFormer-B0.

3.4 AAMO: Average Attention–Mask Overlap

Standard practice evaluates attention maps only qualitatively, by visual inspection of heatmaps. We instead define AAMO to make attention fidelity reportable and comparable the same way Dice or IoU are. Given normalized attention $A \in [0, 1]^{256 \times 256}$ and ground-truth mask Y , AAMO is the fraction of forest pixels covered by thresholded attention:

$$A_{\text{thr}} = \mathbb{I}[A > \tau], \quad \tau = 0.5 \quad (7)$$

$$\text{AAMO} = \frac{|A_{\text{thr}} \cap Y|}{|Y|} \quad (8)$$

i.e. AAMO is the recall of the ground-truth forest region under thresholded attention: 1.0 means attention fully covers the forest mask, 0.0 means no overlap. We additionally report $\text{AAMO}_{\text{dice}}$ and AAMO_{IoU} (Dice and IoU between A_{thr} and Y) as symmetric overlap checks, since AAMO alone does not penalize attention that also covers large non-forest regions.

4 Experimental Setup

Dataset. 5,108 RGB aerial images (256×256) with binary forest/non-forest masks; 70/15/15 stratified train/val/test split. A full dataset audit confirms 0 orphaned image/mask pairs, 0 unreadable files,

and that masks are already effectively binary — no pixel sits more than 9 grey levels from pure black or white, and the mid-gray band (32–223) is empty. The dataset is moderately forest-majority (mean forest-pixel ratio 0.6117 ± 0.3306 per mask), which we account for by reporting Dice/IoU/F1 rather than pixel accuracy alone, since accuracy alone would be inflated by the majority class. Preliminary results below use a 400/60/60 CPU-scale subset, not the full 5,108 images.

Baselines. U-Net (CNN) and vanilla SegFormer-B0 (no attention loss), isolating the effect of the proposed attention supervision, plus DeepLabV3+ (MobileNetV3 backbone) as an additional optional CNN reference point.

Metrics. Dice, IoU, F1 (segmentation quality); parameter count, GFLOPs, FPS (efficiency); AAMO — normalized overlap between thresholded attention and the ground-truth forest mask (interpretability).

Implementation details. SegFormer-B0 variants are trained with AdamW ($\text{lr} = 6 \times 10^{-5}$), batch size 4 (vanilla) / 1 (attention variant, due to the double-backpropagation memory cost), Gaussian target smoothing $\sigma = 8$, $\lambda_2 = 0.3$, and L_{att} using the MSE formulation, seed 42. Both the MSE and KL-divergence formulations of L_{att} are implemented and unit-tested, but only MSE has been evaluated on trained checkpoints so far. All code runs CPU-only in this preliminary phase; the full-scale run uses an identical training loop on a GPU runtime, changing only dataset size and epoch count.

Reproducibility. The attention-extraction and loss implementation is covered by 13 unit tests independent of any trained model (8 for the Attention Consistency Loss, 5 verifying Grad-Rollout output shapes at each SegFormer stage), and AAMO by 10 further tests on synthetic attention/mask pairs, all passing. Dataset integrity is checked by a standalone auditing script (pairing, geometry, value spread, class balance) that exits non-zero on failure, so it can run as a pre-flight gate before training rather than being assumed correct.

5 Preliminary Results

Table 1 reports an 8-epoch, 400-image CPU smoke run. AAMO increases approximately $30\times$ ($0.0144 \rightarrow 0.432$) with the attention-consistency variant, while Dice and IoU also improve slightly over vanilla SegFormer-B0, suggesting the attention supervision does not cost segmentation accuracy at this scale. We additionally report DeepLabV3+ (MobileNetV3 backbone) as an optional CNN reference point beyond U-Net; on a 200-image smoke run it trails both U-Net and SegFormer-B0 on Dice/IoU and shows a strong precision/recall imbalance (precision 0.59, recall 0.98), indicating a bias toward over-predicting forest rather than delineating boundaries precisely.

Limitations of these preliminary numbers. Eight epochs on 400 images is a smoke-scale proof that the pipeline is correct end-to-end, not a reliable trend; the full 5,108-image, 20-epoch run is pending. Figure 2 shows the qualitative attention-drift comparison does not yet tell as clean a story as the AAMO number: the vanilla model’s attention exhibits a corner artifact rather than the motivating roads/shadows failure mode, plausibly from undertraining, and warrants revisiting once the full-scale model has properly converged.

Table 1: Preliminary comparison (CPU/local smoke runs; SegFormer variants: 400/60/60 images, 8 epochs; DeepLabV3+: 200-image smoke run, optional extra baseline).

Model	Dice	IoU	F1	AAMO
U-Net (CNN)	0.8492	0.7379	0.8492	n/a
DeepLabV3+ (MobileNetV3)	0.7369	0.5834	0.7369	n/a
SegFormer-B0	0.8017	0.6690	0.8017	0.0144
SegFormer-B0 + L_{att}	0.8191	0.6936	0.8191	0.432

6 Discussion

The large AAMO gain relative to a small Dice/IoU change suggests attention faithfulness and segmentation accuracy are, at least partially, separable objectives at this scale — supervising attention does not appear to trade off against accuracy, though this must be confirmed with the full-scale run and a multi-seed ablation study before drawing firm conclusions. Both the MSE and KL-divergence formulations of L_{att} are implemented; only MSE has been evaluated so far.

Threats to validity. (1) *Scale.* Results use 7.8% of the dataset (400/5,108 images) for 8 epochs; the direction of the AAMO gain is unlikely to reverse, but its magnitude may shift once the model is properly converged. (2) *Single seed.* No run has been repeated with a different seed, so the reported numbers carry unknown variance; Phase 2’s multi-seed ablation is required before any number here is treated as final. (3) *CNN baseline provenance.* The reported U-Net baseline’s training protocol is still being reconciled against the SegFormer runs’ shared split and training loop, and may be revised. (4) *Rollout restriction.* Restricting Gradient-weighted Attention Rollout to SegFormer’s stage-4 attention avoids the invalid rectangular-matrix case but discards information from stages 1–3, which this work does not yet quantify the cost of.

Broader impact. Forest-cover monitoring increasingly informs deforestation-tracking and conservation-policy decisions, where an automated model’s *explanation* of a prediction — not just its accuracy — affects whether domain experts and policymakers can trust it. An attention map that reliably attends to canopy rather than roads or shadows is a step toward auditable forest monitoring, but these preliminary, small-scale, single-seed results are not yet reliable enough to inform real deforestation-monitoring decisions.

7 Conclusion and Future Work

We presented an explainability-guided SegFormer-B0 that supervises internal attention against the ground-truth forest mask via an Attention Consistency Loss, an adaptation of Gradient-weighted Attention Rollout to SegFormer’s spatial-reduction attention, and a new quantitative interpretability metric, AAMO. Preliminary small-scale results support the core hypothesis. Future work will run the complete multi-seed ablation study at full scale, compare the MSE and KL-divergence loss formulations, tune λ_2 via a validation sweep, and, time permitting, add a boundary-refinement module targeting canopy-edge errors.

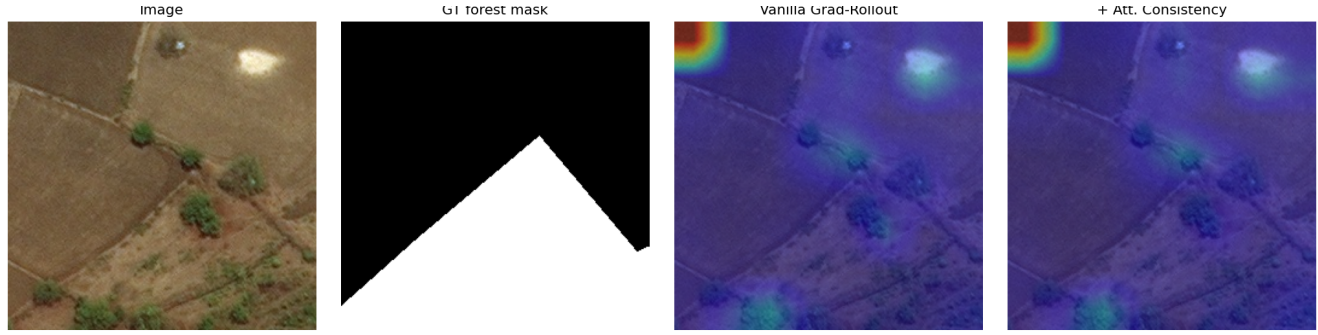


Figure 2: Qualitative comparison on a held-out test image (left to right): input image; ground-truth forest mask; vanilla SegFormer-B0 attention (Grad-Rollout); attention after adding the Attention Consistency Loss. The attention-consistency map is visibly broader and warmer over more of the image, consistent with the large AAMO gain, but does not yet cleanly separate canopy from non-forest by eye; the vanilla map is dominated by a corner artifact rather than the motivating roads/shadows failure mode. Genuine output of the trained checkpoints, not cherry-picked, from an 8-epoch/400-image smoke run.

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